Multimedia Signal Processing - 2nd module

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1 Question - 10 pts

Consider the sequence of symbols, emitted by a continuous source with memory

$$x = \{1, 2, 2, 3, 4, 4\} \tag{1}$$

- Neglecting the source memory
  - Design a Lloyd-Max quantizer with $R = 1$ bit/symbol (assume the initial quantization levels $y_1 = 1$ and $y_2 = 2$ and perform two iterations).
  - Compute the quantized sequence of symbols $\tilde{x}_{dms}$.
  - Compute the quantization SNR_{dms}.
  - Compute the rate R_{dms} when entropy coding is enabled at the output of the quantizer.
- Considering the source memory
  - Implement a method of your choice to take memory into account, by allocating $R_{mem} =$ 1 bit / symbol.
  - Compute the quantized sequence $\tilde{x}_{mem}$.
  - Compute the quantization SNR_{dms}.

In both cases, you can safely assume that x is a representative sample of a stationary source (in practice, you might need a much larger number of samples). Plot the operational rate-distortion points obtained above in a SNR vs. R diagram.

2 Question - 5 pts

Describe the process of motion estimation by block matching, specifying parameters, inputs, outputs and the pseudocode of the algorithm.
Briefly describe the effects of the tunable parameters in terms of accuracy and computational complexity.

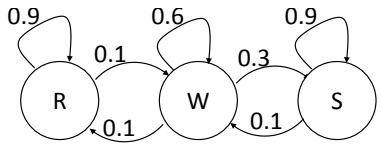
3 Questions - 5 pts (each answer can be either TRUE or FALSE)

1. Consider a transform coding system that adopts blocks of N symbols. When $N = 2$, let σ_1^2 and σ_2^2 denote, respectively, the variance of the first and second symbol in each block.
   - [T] [F] If $\sigma_1^2 > \sigma_2^2$, then under optimal bit allocation $R_1 \geq R_2$
   - [T] [F] If $\sigma_1^2 = 1$ and $\sigma_2^2 = 2$, the coding gain is 1.0607
   - [T] [F] The coding gain is monotonically non-increasing in the block length N
2. Consider a speech coding system
   - [T] [F] Coding intelligible speech typically requires at least 32 kbps
   - [T] [F] The perceptual weighting filter emphasizes the error in the formant regions of the spectrum
   - [T] [F] Voiced speech can be synthesized based on the spectral envelope only
   - [T] [F] The computational complexity of the encoder in A-DPCM is smaller than in analysis-by-synthesis vocoders
3. Consider a block-based image coding system
   - [T] [F] Coding of color images is carried out in the RGB color space.
   - [T] [F] Quantization exploits inter-pixel correlation.
   - [T] [F] In JPEG, the user is able to set the desired target bitrate.

4 Question - 12 pts

A wireless sensor needs to communicate to the base station the state of the processes running on its CPU. Each process can be either in run (R), wait (W) or sleep (S) state. The Figure below illustrates the transition probabilities $p(s_t|s_{t-1})$. The marginal probabilities can be determined to be

$$p(s_t = R) = 0.2, \quad p(s_t = W) = 0.2, \quad p(s_t = S) = 0.6$$



- Implement a Huffman code, encoding blocks of $N = 3$ and symbols assuming a discrete memoryless source.
- Implement the Elias coding algorithm, taking into account the source memory (if you are unable to cope with this problem, assume a discrete memoryless source).

For the sequence of symbols $SSWR$:

1. Encode the sequence
2. Display the output bitstream
3. Decode the sequence

Repeat the steps above for both methods, indicating the number of bits spent by each of them.